Online JEPA Composite Stability: A Two-Time-Scale Extension of Borkar (2008) for Recursive Controlled Systems with Learned Substrates

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***Abstract*—*Scaffold placeholder.* Paper 0 of the Recursive Controlled Systems series proves a per-level stability budget $\lambda_k = \gamma_k - L_\theta \rho_k - L_d \eta_k - \beta \bar{\tau}_k - \frac{\ln \nu}{\tau_a}_k$ under hand-set Lyapunov functions and hand-set constants. This paper generalises the budget to the setting in which the substrate V_k itself is *learned online* by a Joint-Embedding Predictive Architecture (JEPA), yielding two new cost terms: an *online encoder cost* $L_{o\theta}(\Delta, \mu)$ derived by extending Borkar's Ch. 6 two-time-scale stochastic approximation to discrete deployment events, and a *head cost* $L_H\,\kappa_k$ that bounds substrate sensitivity to head-specific payload variations. The new bound reduces to Paper 0's Theorem VI exactly when $\Delta \to \infty$ (frozen substrate) and $L_H \leq 1$ (non-amplifying head); otherwise it is strictly tighter and gates a CI-enforced production rollout via the empirical-constants protocol of Section VI. The final abstract will be written in T5 once the constants table from Q3 substrate measurements is in hand.**

***Index Terms*—stability budget, recursive controlled systems, joint-embedding predictive architecture, two-time-scale stochastic approximation, learned Lyapunov function, deployment-lag analysis, autonomous agents**

I. Introduction

II. Related Work

III. RCS Recap

IV. Theorem Statement

V. Proof

VI. Empirical-Constants Estimation Protocol

VII. Discussion

VIII. Conclusion

References

[1] V. S. Borkar, *Stochastic Approximation: A Dynamical Systems Viewpoint*. Cambridge University Press / Hindustan Book Agency, 2008, ch. 6 is the foundational two-time-scale stochastic approximation reference; extended by Paper 5 T1 (BRO-989) to discrete deployment events.

[2] J. N. Tsitsiklis and B. Van Roy, "An analysis of temporal-difference learning with function approximation," *IEEE Transactions on Automatic Control*, vol. 42, no. 5, pp. 674–690, 1997.

[3] C. M. Bishop, *Pattern Recognition and Machine Learning*, ser. Information Science and Statistics. Springer, 2006, ch. 10 (variational inference) is the standard reference for the KL-bound lemma supporting Paper 5 (H8d).